

Question_6:

Code:

```
1  /*
2   Question No: 6
3   Maintain an array with two operations.
4   1 a b -> add 1,2,3,... to the elements from a to b.
5   2 a b -> print the sum from a to b.
6   A Segment Tree is used.
7   */
8  #include <iostream>
9  #include <vector>
10 using namespace std;
11
12 long long sum[400005], lazyA[400005], lazyB[400005];
13 void build(int k, int l, int r){
14     if(l==r){
15         cin>>sum[k];
16         return;
17     }
18     int m=(l+r)/2;
19     build(2*k, l, m);
20     build(2*k+1, m+1, r);
21     sum[k]=sum[2*k]+sum[2*k+1];
22 }
23 void push(int k, int l, int r){
24     if(!lazyA[k]&&!lazyB[k]) return;
25
26     int m=(l+r)/2;
27     for(int x:{2*k, 2*k+1}){
28         int L=(x==2*k?l:m+1);
29         int R=(x==2*k?m:r);
30         int len=R-L+1;
31         sum[x]+=lazyA[k]*(L+R)*len/2+lazyB[k]*len;
32         lazyA[x]+=lazyA[k];
33         lazyB[x]+=lazyB[k];
34     }
35     lazyA[k]=lazyB[k]=0;
36 }
37 void update(int k, int l, int r, int ql, int qr, int A, int B){
38     if(r<ql||l>qr) return;
39
40     if(ql<=l&&r<=qr){
41         sum[k]+=A*(l+r)*(r-l+1)/2+B*(r-l+1);
42         lazyA[k]+=A;
43         lazyB[k]+=B;
44         return;
45     }
```

```
46     push(k,l,r);
47     int m=(l+r)/2;
48     update(2*k,l,m,ql,qr,A,B);
49     update(2*k+1,m+1,r,ql,qr,A,B);
50
51     sum[k]=sum[2*k]+sum[2*k+1];
52 }
53 long long query(int k,int l,int r,int ql,int qr){
54     if(r<ql||l>qr) return 0;
55
56     if(ql<=l&&r<=qr) return sum[k];
57
58     push(k,l,r);
59     int m=(l+r)/2;
60
61     return query(2*k,l,m,ql,qr)+
62            query(2*k+1,m+1,r,ql,qr);
63 }
64
65 int main(){
66     int n,q;
67     cin>>n>>q;
68     build(1,1,n);
69
70     while(q--){
71         int t,a,b;
72         cin>>t>>a>>b;
73         if(t==1)
74             update(1,1,n,a,b,1,1-a);
75         else
76             cout<<query(1,1,n,a,b)<<"\n";
77     }
78 }
```

Output:

Test Case_1:

```
5 3
3 7 2 2 5
3 2 3
9
1 3 6
1 2 3
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
5 3
4 2 3 1 7
2 1 5
17
1 1 5
2 1 5
32
PS C:\Users\Prabin Yadav\
```